$\pi^{3.14}$: Training-Free Acceleration for Visual Geometry Reconstruction

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Abstract

Feed-forward models have achieved strong performance for visual geometry reconstruction, but their scalability to larger scenes is limited by the quadratic cost of global attention mechanisms. Previous methods like FastVGGT demonstrate that token merging is effective in significantly reducing inference time, but their effectiveness under weaker inductive biases is unclear. In this work, we investigate training-free token merging schemes under permutation-equivariant architecture and introduce $\pi^{3.14}$, a token merging scheme that uses KMedoids to cluster input frames to significantly reduce sequence length. Experiments on Sintel and NRGBD show degradation in reconstruction quality, but consistent speedups in inference time for larger scenes. These results highlight both the challenges and promise of token merging for visual geometry reconstruction. Code is available at: Pi3.14.

1. Introduction

Visual geometry reconstruction, the process of creating 3D models from visual inputs, is a crucial step towards allowing machines to perceive the world and surrounding environment for further downstream tasks like augmented reality and autonomous devices. State of the art works have evolved from iterative optimization techniques like Bundle Adjustment (Hartley & Zisserman, 2003) to end-to-end feed-forward models like VGGT (Wang et al., 2025a) and π^3 (Wang et al., 2025c) which enable reconstructing scenes from multiple views.

These latest feed-forward reconstruction models adopt a transformer-based architecture, with the π^3 (Wang et al., 2025c) model but ditching the long-standing inductive bias of selecting a single fixed reference view. Although eliminating this inductive bias does improve model accuracy, it does not address the computational cost of attention that is quadratic in sequence length. FastVGGT (Shen et al., 2025) provides a inference time profile of the VGGT (shown in Figure 1) model which shows how the Global Attention module (attention across all input frames) quickly dominates the inference time. Consequently, introducing optimizations to the Global Attention module will see the most benefit to performance.

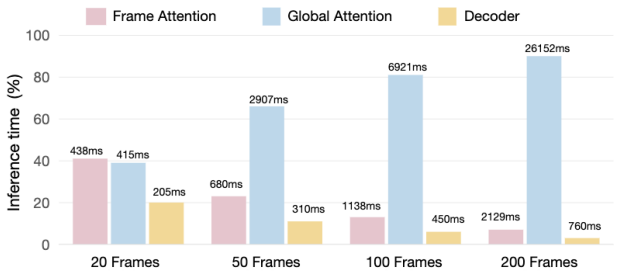


Figure 1. Component-wise analysis of VGGT inference time, taken from the FastVGGT paper.

It is important to note that the π^3 model is not the same but shares a very similar architecture to the VGGT model, so we will assume that similar findings for runtime profiling also hold and motivate this research. Hence, this paper continues along the path of exploring training-free optimizations, namely, through token merging. Training-free methods were chosen to avoid the expensive computational costs of training and have the potential to be portable across different models. The token merging schemes consist of partitioning the tokens into a set of source (src) and destination (dst) tokens, merging the src and dst tokens, and finally unmerging the tokens after calculating attention scores. Two implementations of were tested: one baseline using region-based random sampling similar to that used in ToMeSD (Bolya & Hoffman, 2023) and based on FastVGGT's scheme (Shen et al., 2025), and we introduce the second as $\pi^{3.14}$, a novel approach to token merging by clustering patch tokens across all frames.

In summary, the main contribution of this work is exploring whether training-free token merging techniques are still performant under a weaker inductive bias. Two token merging schemes are tested: 1) adapting a token merging scheme from a prior work and 2) $\pi^{3.14}$, a novel merging scheme

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that investigates the effectiveness of utilizing a more intelligent partitioning algorithm. Although experiments show there is a degradation in model performance, the reduction in inference time for large scenes show the promise of token merging with more informed partitioning schemes and/or integration into the pre-training stage.

2. Related Work

2.1. 3D Reconstruction

Traditional 3D reconstruction techniques utilize feature correspondences between images to estimate camera poses and create dense 3D point clouds. Although methods like Structure-from-Motion (SfM) (Hartley & Zisserman, 2003; Schonberger & Frahm, 2016) and Multi-View Stereo (Furukawa & Hernández, 2015; Schönberger et al., 2016) are robust, they rely on complex pipelines and time-consuming iterative optimization techniques like Bundle Adjustment to jointly refine 3D points and camera poses.

More recent works have employed end-to-end feed-forward networks to predict the structure of a scene from a set of images without needing the aforementioned time-consuming techniques. Dust3R (Wang et al., 2024) was one of the seminal models which processed image pairs to predict point clouds, but required a global alignment step for processing larger scenes. To avoid this time-consuming and fragile process, works like Fast3R (Yang et al., 2025) marked a new era of 3D reconstruction models which could perform inference on thousand of images simulatenously. VGGT (Wang et al., 2025a) expands upon these works by estimating all 3D attributes of a scene including camera poses, depth maps, point maps, and 3D point tracks, rather than solely predicting point clouds.

One key constraint among all these approaches is an unnecessary inductive bias that was inherited from SfM techniques which is the reliance on a fixed reference view for predicting the 3D structure. The current state-of-the-art work $\boldsymbol{\pi^3}$ (Wang et al., 2025c) eschews all order-dependent components like positional embeddings to differentiate between frames and learnable camera tokens that designate a reference view. By omitting these order-dependent components, $\boldsymbol{\pi^3}$ is able to achieve permutation-equivariance, improving robustness to input ordering and achieving improved performance on a wide range of tasks. However, this state-of-the-art still suffers from large computational overhead from the global attention mechanism which computes attention across all frames. To overcome this, we propose $\boldsymbol{\pi^{3.14}}$, which aims to accelerate inference by reducing the sequence length in Global Attention, in this new setting of permutation-equivariant architecture.

2.2. Token Merging

Vision Transformers saw the first application of token merging (Bolya & Hoffman, 2023) as a training-free way to improve throughput. Naturally, this approach was adopted for other image and video understanding tasks (Cao et al., 2023; Shen et al., 2024). Token merging involves partitioning the tokens into two sets, matching them based on similarity, and merges these redundant tokens by averaging. However, this technique is limited in application to images but applications into long-form video are underexplored. FastVGGT (Shen et al., 2025) introduces token merging to the feed-forward 3D reconstruction and achieves substantial acceleration while maintaining reconstruction quality, despite the unique challenges of retaining spatial and temporal consistency. However, this recent approach is applied to a model architecture which differs from that of the state-of-the-art work of $\boldsymbol{\pi^3}$ and lacks interpretability. To address these, we propose $\boldsymbol{\pi^{3.14}}$, which adapts token merging to this permutation-equivariant architecture and implements a more interpretable partitioning scheme based on frame clustering.

3. Method

3.1. Motivation

Following the observations made in FastVGGT (Shen et al., 2025), targeting Global Attention would yield the greatest speedup. To achieve this speedup, we explore a new partitioning scheme which clusters similar frames. The idea of matching features was present in traditional 3D reconstruction techniques, but were replaced by more efficient methods that regress in a single forward pass. However, rather than being a core part of the algorithm for predicting attributes of a 3D scene, $\boldsymbol{\pi^{3.14}}$ leverages these shared features to reduce the number of tokens processed. Frames share a lot of redundant information, especially when derived from video, so clustering similar frames is a valid approach to reduce the number of frames processed while still retaining important scene information. Visual image tokens also require a lot more context so reducing the number of frames via clustering can improve scaling to much larger scenes.

3.2. Token Partitioning

As mentioned previously, frames of a particular scene may contain redundant information that is shared across viewpoints. $\boldsymbol{\pi^{3.14}}$ already encodes input frames using a DINOv2 (Oquab et al., 2024) backbone. Using these features, we first average these patch features for each image, yielding N descriptors, one for each input frame. These descriptors are used as inputs for the KMedoids (Maranzana, 1963; Park & Jun, 2009) algorithm which selects k with $k \ll N$ descrip-

tors to act as the representative image tokens. The original patch tokens belonging to these k descriptors are marked as destination tokens and the rest are marked as source tokens.

3.3. Token Merging Procedure

After partitioning, each source patch token x_s is assigned to the patch token x_d of the corresponding cluster center (one of the aforementioned k descriptors) and is updated by averaging:

$$x_{d'} = \frac{x_d + x_s}{2}$$

The updated patch token $x_{d'}$ is retained for the attention calculations while the source x_s is discarded, thus reducing the number of tokens processed.

3.4. Token Unmerging Procedure

Dense 3D reconstruction requires full output resolution so we employ an unmerging operation that is inspired by ToMeSD (Bolya & Hoffman, 2023). Suppose $x_1, x_2 \in \mathbb{R}^D$ were previously merged via

$$x^*_{1,2} = \frac{x_1 + x_2}{2}$$

Then, during unmerging, the merged result is simply replicated, i.e.

$$x'_1 = x^*_{1,2},\ x'_2 = x^*_{1,2}$$

This ensures matching input and output resolutions of the attention module and compatibility with the decoder to make predictions for each patch.

4. Experiments

4.1. Experimental Setup

Due to limited computation resources, $\boldsymbol{\pi^{3.14}}$ was evaluated on two datasets: Sintel and NRGBD. We focus on two tasks: camera pose estimation and point map estimation on Sintel and NRGBD, respectively. We test the original $\boldsymbol{\pi^3}$ model, and an adapted implementation of FastVGGT and $\boldsymbol{\pi^{3.14}}$ with different merge ratios r, which determines the percentage of tokens to reduce the input sequence by. The overall architecture is the exact same as $\boldsymbol{\pi^3}$. All experiments were done on a laptop with an RTX 3060 Laptop GPU (6 GB VRAM).

4.2. Camera Pose Estimation

For camera pose estimation, the predicted camera pose was assessed using distance error metrics. Following (Wang et al., 2025c), we report the Absolute Trajectory Error (ATE), Relative Pose Error for translation (RPE-t), and Relative Pose Error for rotation (RPE-r) on the Sintel (Azinović et al., 2022) dataset. We also include the average time taken for inference. As shown in Table 1, the results show that our method does not perform well, but is successful in reducing inference time.

	ATE ↓	RPE-t ↓	RPE-r ↓	Time (s) ↓
$\boldsymbol{\pi^3}$	<u>0.125</u>	**0.041**	**0.294**	384.871
Baseline ($r = 0.3$)	0.587	0.353	4.3142	289.368
Baseline ($r = 0.6$)	0.256	0.174	1.159	231.316
Baseline ($r = 0.9$)	**0.121**	<u>0.058</u>	<u>0.443</u>	227.268
$\boldsymbol{\pi^{3.14}}$ ($r = 0.3$)	0.660	0.244	4.631	252.742
$\boldsymbol{\pi^{3.14}}$ ($r = 0.6$)	0.649	0.212	3.789	203.870
$\boldsymbol{\pi^{3.14}}$ ($r = 0.9$)	0.595	0.279	3.010	196.516

Table 1. **Camera pose estimation** on **Sintel** dataset. r defines the merging ratio.

4.3. Point Map Estimation

For point map estimation, we evaluate the quality of the reconstructed scenes on the NRGBD (Azinović et al., 2022) dataset under two different sampling strides (denoted sparse and dense). Following prior works (Azinović et al., 2022; Wang et al., 2024; Wang & Agapito, 2025; Wang et al., 2025b), we report the Accuracy (Acc.), Completion (Comp.), and Normal Consistency (N.C.). We also include the average time taken for inference. As shown in Table 2, the results show that our model still does not perform well for this task, but is able to reduce inference time for larger scenes.

	View	Acc.↓		Comp.↓		NC.↑		Time(s) ↓
		Mean	Med.	Mean	Med.	Mean	Med.	
$\boldsymbol{\pi^3}$		**0.030**	**0.018**	**0.032**	**0.016**	**0.908**	**0.994**	20.603
Baseline ($r = 0.3$)		0.224	0.181	0.549	0.440	0.604	0.649	21.396
Baseline ($r = 0.6$)		0.240	0.187	0.394	0.321	0.656	0.742	21.758
Baseline ($r = 0.9$)	sparse	<u>0.062</u>	<u>0.039</u>	<u>0.084</u>	<u>0.050</u>	<u>0.829</u>	<u>0.963</u>	21.872
$\boldsymbol{\pi^{3.14}}$ ($r = 0.3$)		0.250	0.227	0.678	0.631	0.625	0.674	21.852
$\boldsymbol{\pi^{3.14}}$ ($r = 0.6$)		0.240	0.198	0.774	0.667	0.624	0.665	22.345
$\boldsymbol{\pi^{3.14}}$ ($r = 0.9$)		0.246	0.206	0.726	0.627	0.632	0.686	22.474
$\boldsymbol{\pi^3}$		**0.012**	**0.007**	**0.014**	**0.006**	**0.883**	**0.986**	140.167
Baseline ($r = 0.3$)		0.233	0.209	0.730	0.575	0.580	0.628	124.048
Baseline ($r = 0.6$)		0.166	0.116	0.440	0.368	0.635	0.723	108.145
Baseline ($r = 0.9$)	dense	<u>0.037</u>	<u>0.025</u>	<u>0.030</u>	<u>0.013</u>	<u>0.815</u>	<u>0.957</u>	102.612
$\boldsymbol{\pi^{3.14}}$ ($r = 0.3$)		0.268	0.241	0.785	0.651	0.581	0.622	112.408
$\boldsymbol{\pi^{3.14}}$ ($r = 0.6$)		0.230	0.208	0.755	0.613	0.580	0.614	100.113
$\boldsymbol{\pi^{3.14}}$ ($r = 0.9$)		0.248	0.225	0.758	0.596	0.577	0.616	90.964

Table 2. **Point map estimation** on **Neural-NRGBD** dataset. r defines the merging ratio.

5. Discussion

Although $\boldsymbol{\pi^{3.14}}$ yields a degradation in model performance, this work does provide insights into the application of token merging to visual geometry reconstruction.

Adapting to permutation equivariant architecture. In experiments across both datasets, we see that simply adapting FastVGGT to $\boldsymbol{\pi^3}$ yields significant speedups while achieving very similar model performance. This makes sense

as there aren't any drastic architectural changes between VGGT and π^3 and the camera token in VGGT was simply replaced with another register token to encode global information. Nevertheless, we see that training-free token merging can still be applied to the state-of-the-art work that ditches order-dependent components.

Applicability for smaller scenes. In the point map estimation experiments, we see that although FastVGGT token merging performs similarly to the original model, it comes at the cost of slower inference for smaller scenes (fewer input frames). Inherently, there is a tradeoff between the speedup from reducing the number of frames processed and the computational overhead incurred by the merging strategy. For smaller scenes, it makes sense that this overhead is greater due to additional operations from the partitioning, merging, and unmerging schemes, all while reducing the token count in a limited capacity. Token merging may not necessarily be a free lunch and is detrimental for quick inference on smaller input sizes.

Tradeoff in merge ratio. Across both experiments, we see the adapted FastVGGT implementation suffer with smaller merge ratios but then performance improves with significantly higher merge ratios. Naturally, there is a loss of information through the merging procedure and the merge ratio, r, controls this tradeoff. While larger merge ratios significantly reduce the number of tokens processed, it may result in a degradation of model performance through the loss of important information. On the other hand, smaller merge ratios may include too much redundant information that also adds noise when merged together, which also reduces model performance. Hence, optimal merging ratios need to be finetuned for the task at hand otherwise downstream task performance may suffer.

Inference time speedup. For larger scenes, the experiments demonstrate significant inference time speedups which align with our intuition that reducing the input size into the global attention module will accelerate inference. While $\pi^{3.14}$ suffers in degraded model performance, the motivation behind the approach is well-founded and is shown to speedup inference, potentially even more than the naive approach. A training-free approach which realizes this intuition and manages to also maintain model performance is an area for future work. The significant decrease in inference time can offset the additional overhead incurred by more intelligent frame clustering schemes and more interpretable approaches so that overall inference time is still reduced compared to the base model. Moreover, with this insight in mind, newer models could adopt token merging as part of the pre-training stage so that the model learns how to work directly with these compact representations from merging.

6. Conclusion

In this work, we investigated token merging as a training-free method to accelerate visual geometry reconstruction models with permutation-equivariant architecture. In addition, we propose $\pi^{3.14}$, a token merging scheme that exploits redundant features shared across viewpoints and clusters frames with the goal of reducing inference time while preserving model performance. While this work was unsuccessful at achieving similar model performance to the base model and a baseline token merging scheme, it has provided insights into the future of token merging for visual geometry reconstruction. It demonstrates that 1) simple and existing token merging schemes still apply to permutation-equivariant architecture, 2) downstream tasks requiring finetuning to find optimal merging ratios, and 3) inference time may suffer in smaller scenes, but yields significant speedups for larger scenes and more intelligent token merging schemes and/or integration into the pre-training stage can better take advantage of redundant information across multiple viewpoints.

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